Collecting Numbers II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 2217 | Sorting and Searching | https://cses.fi/problemset/task/2217

You are given an array that contains each number between 1 … n exactly once. Your task is to collect the numbers from 1 to n in increasing order.

On each round, you go through the array from left to right and collect as many numbers as possible.

Given m operations that swap two numbers in the array, your task is to report the number of rounds after each operation.

Input

The first line has two integers n and m: the array size and the number of operations.

The next line has n integers $x_1,x_2,\ldots,x_n$: the numbers in the array.

Finally, there are m lines that describe the operations. Each line has two integers a and b: the numbers at positions a and b are swapped.

Output

Print m integers: the number of rounds after each swap.

Constraints

- $1 \le n, m \le 2 \cdot 10^5$
- $1 \le a,b \le n$

Example

Input

```
5 3
4 2 1 5 3
2 3
1 5
2 3
```

Output

```
2
3
4
```